

Wanli Qian

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Website | LinkedIn | GitHub | Google Scholar

Seeking Summer 2027 Research Internship

Robotics | Tactile Perception | Teleoperation

Research Profile

CS Ph.D. researcher studying how robots acquire and represent physical properties through touch, multimodal sensing, and action. Builds tactile-feedback teleoperation systems and language-conditioned haptic models. Research spans active perception, manipulation, and human-robot interaction; recipient of a 2026 IEEE Haptics Symposium paper award.

Education

University of Southern California

Ph.D. in Computer Science

2024–Present

Expected graduation: Fall 2028

University of Chicago

Predoctoral M.S. in Computer Science

2022–2024

Georgia Institute of Technology

B.S. in Computer Science

2018–2022

Research Experience

USC HaRVI Lab | Ph.D. Researcher

2024–Present

- Build bilateral teleoperation systems with tactile feedback for remote surface perception, manipulation, and data collection.
- Study active haptic perception through exploratory policies that adapt robot actions to reduce uncertainty about object and surface properties.
- Develop language-conditioned generative models for tactile and vibration signals using force, acceleration, and vision-based tactile data.
- Curate multimodal haptic datasets spanning 200+ materials, combining tapping, sliding, audio, and affective descriptors.
- Integrate VR/AR and language-based interaction into haptic and texture authoring systems.

University of Chicago AxLab | Predoctoral Researcher

2022–2024

- Led the development of SHAPE-IT into an interactive system for conversational 3D shape generation and real-time reconfiguration of physical shape displays.
- Developed adaptive LLM-based agents that modify shape-display control scripts in response to ongoing interaction.

Georgia Tech Borg Lab | Lab Assistant

2021–2022

- Developed motion planning and trajectory optimization methods for robotic drawing and writing with cable robots and articulated manipulators.
- Integrated motion capture with robot control to analyze and reproduce human artistic trajectories; explored polynomial stroke models and deformable virtual brush dynamics.

Industry Experience

Kolmostar | Software Engineer Intern

2022

- Built urban-geometry simulation pipelines to quantify wireless-signal shadowing, with visualization tools supporting sensing-system deployment.

Megvii | Robotics Software Solution Intern

2019

- Implemented and benchmarked probabilistic and graph-based motion planners for high-DOF industrial manipulators; visualized efficiency, robustness, and convergence across task geometries.

Selected Publications

Language-Guided Multimodal Texture Authoring via Generative Models.

Wanli Qian, Michael Gu, Aiden Chang, Shihan Lu, Heather Culbertson.

IEEE Haptics Symposium, 2026. [Paper](#) | [Video](#)

Distinguished Technical Long Paper Award

SHAPE-IT: Exploring Text-to-Shape-Display for Generative Shape-Changing Behaviors with LLMs.

Wanli Qian*, Chenfeng Gao*, Anup Sathya, Ryo Suzuki, Ken Nakagaki.

ACM UIST, 2024. *Equal contribution. [Paper](#) | [Video](#)

Shape n'Swarm: Hands-on, Shape-aware Generative Authoring with Swarm UI and LLMs.

Matthew Jeung, Anup Sathya, Wanli Qian, Steven Arellano, Luke Jimenez, Ken Nakagaki.

ACM UIST, 2025. [Paper](#) | [Video](#)

GTGraffiti: Spray Painting Graffiti Art from Human Painting Motions with a Cable Driven Parallel Robot.

Gerry Chen, Sereym Baek, JD Florez, Wanli Qian, Sang-won Leigh, Seth Hutchinson, Frank Dellaert.

IEEE ICRA, 2022. [Paper](#) | [Video](#)

Weakly Supervised Part-Based Method for Combined Object Detection in Remote Sensing Imagery.

Wanli Qian, Zhe Yan, Zhe Zhu, Wei Yin.

IEEE JSTARS, vol. 15, pp. 5024–5036, 2022. [Paper](#)

Manuscript Under Review

TouchTwin: Human–AI Haptic Authoring for 3D-Scanned Tabletop Scenes through Language and Touch.

Wanli Qian et al.

Submitted to ACM CHI 2027; under review.

Teaching

CSCI 445: Introduction to Robotics | Teaching Assistant, USC

2025

CMSC 25040: Introduction to Computer Vision | Teaching Assistant, UChicago

2023

Service & Outreach

Reviewing: CHI, UIST, IEEE Haptics Symposium, IEEE Transactions on Haptics.

Mentorship: USC CURVE (Spring/Fall 2025, Spring/Fall 2026); USC Viterbi SURE (Summer 2025); USC REU (Summer 2026).

Outreach: Peace4Kids robotics workshop organizer/mentor; RSS conference volunteer.